

Computational Reproduction Record

This repository reproduction record contains the complete preceding computational narratives and tables, identified by their original revision, followed by every new exact-pooling and numerical mixed-integer case. It is code-and-experiment documentation, not an additional proof appendix. The current article and electronic companion retain all mathematical statements and proofs. Historical measurements are not relabeled as fresh executions.

CR.1 Computational Evidence for Joint Design

The primary protocol fixes 24 independent-seed synthetic cases with two, four, or eight input branches; nine or seventeen catalog points; two or three installed commands; and zero or nonnegative heterogeneous charges. Eight additional cases vary the raw history count and response-type count using a 33-point catalog. Levels lie in $[0, 1]$, the reward has $r = 2$ and $q = 1$, and shortfall curvatures are drawn from $\{1, 2, 4, 8\}$. Participation caps, weights, and eligibility ceilings vary independently within the declared rational generator. The protocol was frozen before these runs, not externally preregistered. Every generated case is retained. Larger comparator extensions were added after the primary run and cover the entire declared comparison set, not selected successes.

The proposed method uses eight root price-bisection steps, retains price zero, and uses the same budget and charges in every incumbent and bound. Primary box runs have a 301-node and eight-second soft limit; repeated-type scaling runs have a 101-node limit. Time is checked after completing a node, so some recorded times exceed eight seconds. The only initial incumbent is a fixed-target conditional book at the capped ideal targets, followed by exact allocation for that book. It is not a global certificate. The baseline price-prefix search reimplements the inherited R43 exact completion bound, uses the same price budget, and stops at the same absolute tolerance. A weaker full-catalog prefix relaxation is reported separately, not substituted for this stronger baseline.

Table CR.4 reports 28 of 32 box runs attaining absolute gap 0.001, versus 32 of 32 for the priced-prefix baseline. The box method has lower measured optimization time in 11 cases, but it is not uniformly faster or more accurate. Median times are 0.093 seconds for boxes and 0.070 seconds for priced prefixes. Four box runs retain unresolved gaps between 0.00217 and 0.01248. These are valid intervals, not achieved-tolerance claims. The median Jensen-minus-incumbent interval is 0.01124; the median final box interval is 0.00018. Every box interval passes the independent structural checker.

The eight smallest holdout cases also have independent all-book joint references. Their incumbent losses are zero in this sample, but the priced bounds need not be exact. This observation is not a guarantee for the initial target choice or any heuristic portfolio. Repeated-type cases with 128 and 256 histories reduce exactly to four or eight response types; their certificates are lifted and checked on the original history space. The sixteen-type cases have no such large reduction. These rows demonstrate full joint intervals after a stated exact aggregation, not merely the runtime of conditional target compression.

Table CR.5 tests the complete-grid reference at successively smaller tolerances and with two through five branches. Only the three two-branch grid runs complete within the 1,000-profile limit. The adaptive method

completes eleven of twelve corresponding requests; its four-branch, 0.001 run remains interrupted. Exact-reference losses, normalized objective scales, and all counts are provided rather than inferring accuracy from sampled objective agreement. This comparison tests the new coverage mechanism, not a claim that generic spatial partitioning is novel.

The independent mixed-integer formulation retains installation binaries, branch support indicators, probabilities, pre-draw service, the root equality, and every realization implication. Its tangent and secant models are solved on all 24 primary cases with 64 cost segments and separate four-second limits. All 48 solves return optimal numerical status. Table CR.16 gives each case's runtimes, search nodes, primal-dual gaps, envelope error and bracket width. Negative signed widths at floating roundoff scale are retained in the raw records and are not called exact certificates. The results show that generic MIP and inherited prefix search remain competitive on these catalogs. The methodological claim for the box oracle is exact lower-bound-aware decomposition and independently checkable adaptive joint certificates, not universal computational superiority.

Regression includes 600 exact all-book price comparisons on heterogeneous target boxes, 24 further joint interval comparisons, 2,793 independent finite-lattice checks of constructive net coverage, genuine below-anchor and above-cap clipping, the one-branch one-call case, and eleven rejected certificate mutations. The three-target descending-side example in Section 6 is an exact counterexample to reusing a rising-only oracle on a target box. The coverage checker is structurally independent of the search routine. These are verification tests, not statistical evidence about an operational population.

The complete historical numerical study remains in Section CR.5. Its thirty-five-of-thirty-six portfolio result and safeguarded sample exactness are not used as the present global-algorithm headline. Its inserted-ideal-target continuous example is an oracle candidate-pool sanity check, and its large conditional table is explicitly labeled "targets fixed." No customer records, estimated installation charges, field calibration, or deployment outcomes are asserted.

CR.2 Independent Formulation and Numerical Error Bounds

The independent optimizer does not import the canonical response, either path recurrence, or the fixed-book allocation code. Let $z_i \in \{0, 1\}$ enable level a_i , let $w_{ji} \in \{0, 1\}$ indicate a permitted branch support, let $p_{ji} \geq 0$ be its probability, and let $y_j \in [0, b_j]$ be the intermediate tier chosen before the draw. The original quadratic catalog problem is represented by

$$\max \quad \sum_j \pi_j \left[\sum_i p_{ji} f(a_i) - \frac{\gamma_j}{2} y_j^2 \right] - \sum_i \rho_i z_i, \quad (\text{CR.1})$$

$$\text{s.t.} \quad \sum_i z_i \leq s, \quad 0 \leq p_{ji} \leq w_{ji} \leq z_i, \quad \sum_i p_{ji} = 1, \quad (\text{CR.2})$$

$$y_j + \sum_i a_i p_{ji} \leq b_j, \quad \sum_j \pi_j \left(y_j + \sum_i a_i p_{ji} \right) = B, \quad (\text{CR.3})$$

$$y_j + a_i \leq \tau_j + M_{ji}(1 - w_{ji}), \quad M_{ji} = \max\{0, b_j + a_i - \tau_j\}. \quad (\text{CR.4})$$

For $a_i > \tau_j$, set $w_{ji} = p_{ji} = 0$. The bound $y_j \leq b_j$ makes the displayed big- M sufficient. No adjacent-support constraint is imposed. A feasible contract gives a feasible point by activating its positive support and installed levels. Conversely every model point defines one pre-draw tier and a lottery obeying both the target cap and every realized ceiling. Thus the model represents the original feasible set, not its canonical reduction. The allowance of activated levels with zero use does not change the optimum under nonnegative charges: redundant activations can be removed.

For the numerical comparison, replace $\gamma_j y^2/2$ by an epigraph of piecewise-affine functions. With nodes $x_r = r b_j/K$, $0 \leq r \leq K$, the tangent envelope is

$$h_j^{\tan}(y) = \max_{0 \leq r \leq K} \{\gamma_j x_r y - \gamma_j x_r^2/2\}.$$

The secant envelope is the maximum of the K consecutive chord lines. On $[0, b_j]$,

$$0 \leq h_j - h_j^{\tan} \leq \frac{\gamma_j b_j^2}{8K^2}, \quad 0 \leq h_j^{\sec} - h_j \leq \frac{\gamma_j b_j^2}{8K^2}.$$

The tangent formula follows by completing the square and choosing the nearest node; the secant formula follows from the quadratic chord identity. Consequently, writing $E_K = \sum_j \pi_j \gamma_j b_j^2/(8K^2)$, the exact envelope optima satisfy

$$V_s \leq V_s^{\tan} \leq V_s + E_K, \quad V_s - E_K \leq V_s^{\sec} \leq V_s.$$

A secant-model feasible policy is feasible for the original constraints, and evaluating its actual quadratic cost can only improve its objective. In the implementation, however, the solver and its dual bounds are floating point. We therefore retain status, reported mixed-integer gap, residuals, raw probabilities, intermediate tiers, and both envelope results; we do not promote a numerical dual bound to an exact rational certificate. Comparing against exact catalog values is a separate validation step. The chosen numerical tolerances and segment count are stored with every result.

CR.3 Independent Rational Policy Checks

For a fixed target and proposed book, the independent exhaustive routine enumerates singleton and all two-point supports directly. For a singleton $a \leq t$, service is $y = t - a$. For a pair $u < v$, the terminal mean is $t - y$ and the high probability is $(t - y - u)/(v - u)$. Feasible service lies in

$$\max\{0, t - v\} \leq y \leq \min\{t - u, \tau - v\}.$$

Its payoff is the pair's affine reward at $t - y$ minus the nondecreasing intermediate cost. Because the chord slope is positive, the smallest feasible y maximizes this payoff. Enumerating singleton and pair supports therefore gives a direct branch optimum without invoking the canonical implementation. Concavity and the standard two-moment extreme-point argument justify that no larger support is required. Catalog-subset enumeration then gives a separate exact check of conditional book selection.

The policy checker verifies that each activated level belongs to the catalog, the book is strictly ordered and within its stated budget, charges are paid once, each probability is nonnegative, and each lottery sums to one. It recomputes each target as $y_j + \sum_i a_i p_{ji}$, the exact root equality, every cap, and every positive-probability realization $y_j + a_i \leq \tau_j$. It separately recomputes the quadratic reward and intermediate cost. Semantic mutation cases change targets, probabilities, support, service, charges, promise, or the declared budget; the test record identifies each rejection. This checker validates a policy and conditional enumeration. It does not infer global target optimality from a feasible policy or from the same price recurrence.

CR.4 Target Nets, Approximation Errors, and Support Stability

The net implementation fixes a possible lowest installed level, enumerates $k - 1$ weighted target coordinates, fills the remaining coordinate subject to its cap, and repairs any positive residual by bounded increases. It chooses a branch of largest probability as the distinguished coordinate, solely to avoid unnecessary enumeration; the theorem does not depend on this choice. The code stores the weighted mesh step, evaluated profile count, anchor count, requested accuracy, and completion status. A profile cap can interrupt execution, but then the global ε guarantee is explicitly absent. All returned policies still satisfy the resource and contractual constraints.

In floating-point implementations, aggregate equality error alone does not bound all target perturbations. Let t^U be the exact mixed target, and let $\tilde{t}$ be its rounded version after clipping to the chosen book's feasible target intervals. Compute or upper-bound

$$r_{\text{mix}} = \sum_j \pi_j |\tilde{t}_j - t_j^U|, \quad r_{\text{eq}} = \left| B - \sum_j \pi_j \tilde{t}_j \right|.$$

After exact residual repair, the weighted distance from t^U is at most $r_{\text{mix}} + r_{\text{eq}}$. Hence the safe additional value allowance in the compressed-support theorem is $L(r_{\text{mix}} + r_{\text{eq}})$, not merely a bound on the equality residual. Clipping is included in r_{mix} . Canonical reconstruction is performed after repair, and any desired rational probability representation is generated from the repaired targets and exact installed levels. Independently rounding probabilities and then checking only their sum would not establish root or risk feasibility.

The theory does not require continuity of the selected book. Tied optima or perturbed parameters can change several selected levels while the corresponding certified value interval changes little. In particular, uncertain eligibility thresholds must be replaced by declared safe bounds before optimization; a lottery feasible for an estimated threshold is not automatically safe under downward perturbation. The error theorem concerns certified oracle and target arithmetic at stated model data. It is not an unproved robustness result for unknown customer parameters.

CR.5 Current Reproduction and Preserved Evidence

The exact command budget counts selected, enabled levels. It is not the number of levels receiving positive probability after optimization; zero-charge exact-cardinality frontiers may contain unused selections. The

original archival column “Used” is a selected-level count. The current code reports the book and the positive-probability support separately through the full lottery record.

The primary protocol, all original inputs, and completed and interrupted outcomes are retained in `PROTOCOL.json` and `results/`. Run `code/tests.py`, `code/benchmark.py`, `code/comparators.py`, and `code/price_compare.py` to regenerate scientific evidence; `code/tables.py` generates the displayed values. Benchmarks record actual wall time and platform. A resumed run retains completed records rather than selecting new inputs. The comparator extension covers all primary cases, and the priced-prefix extension covers all 32 cases. They were not part of an external preregistration. The separate `check_certificate.py` verifies a supplied rational target-box certificate without importing the search routine.

The complete R45 main paper and electronic companion are preserved in `predecessor/`, alongside their source entry points. Their historical computational sections, protocols, and complete tables remain available there and in their unchanged source directories, rather than being appended again as new experimental evidence. Every mathematical theorem, proposition, lemma, corollary, and proof from the current R45 readers remains in the current article or this companion. The content map identifies each current, moved, or archived input. All inherited repository paths outside the explicitly replaced root readers and index files are byte-preserved.

In the archived charged portfolio exception, the exact same-budget value, the portfolio value and their difference are recorded together in the new response, so the loss 101/3600 is not presented without scale. The archive’s interior target-adaptive exact example includes the ideal targets in its pool; it is an oracle-pool sanity check. Its 32–256-branch timing table optimizes the book conditional on supplied targets and is not pooled with the present joint intervals.

CR.6 Joint Computation with Distinct Histories

The additional protocol contains twelve primary cases with four, eight, or twelve distinct histories and twelve scaling cases with 32, 128, or 256 distinct histories. Primary catalogs have nine levels and budgets two or three; scaling catalogs have seventeen levels and budget three. Four cap centers and four cost centers receive rational, history-specific perturbations at scales 0.001 and 0.02. We vary nonuniform weights, nonnegative charges, one or four eligibility regimes, and interior or saturated promises. Every cap–cost pair is distinct, so the exact response-type reduction leaves every history separate. This is a declared clustered synthetic family, not a generic random population. The protocol and generator hash were fixed before these runs, not externally preregistered.

Coarsening uses cap and curvature bins twice the perturbation scale, exact eligibility classes, and one minimum-cap guard. It gives four groups for four-history cases and five otherwise. These practical bins are deliberately coarser than the conservative accuracy prescription in Theorem EC.15.3; their validity comes from the checked original-space interval, not from assuming a small worst-case error. The inner target-box tolerance is 0.0005, and the requested original tolerance is 0.001. Both box methods use eight root-price steps, a 101-node limit and a four-second soft limit. No original reallocation improvement is added after the groupwise lift. The priced-prefix baseline uses the same original tolerance and resource limits. Soft limits

are checked between complete oracle blocks; recorded times exceeding them are retained.

Tables CR.6 and CR.7 report 24 of 24 original-space certificates within tolerance for coarsening, 19 for unaggregated boxes, and 21 for priced prefixes. Median optimization times are 0.037, 0.318, and 0.169 seconds, respectively. The largest coarsened gap is 0.000783. This is a global certificate result, not an exact-hit claim about a target portfolio. All 48 coarsened and raw intervals pass independent semantic checking; full original policies and tree witnesses are supplied. The inherited 32-case study, including its four unresolved box runs and stronger prefix results, remains intact in Section CR.1. Its runtimes are not pooled with this execution.

All twelve primary cases have exhaustive all-book joint references. That enumeration is independent of the target search but shares the exact fixed-book allocator; it is not presented as a separately derived allocation algorithm. The largest lifted-incumbent loss to those references is $3.03e-13$. A genuinely uneliminated original-model mixed-integer formulation provides a separate check: its 24 tangent/secant solves retain every support implication and do not use canonical allocation. Table CR.14 reports each solver status, time, node count, gap, envelope error, and signed numerical bracket width. These floating-point bounds are not substituted for rational certificates.

An additional four-case stress protocol was fixed after the primary run to challenge coarse recovery, rather than to extend its success rate. In Table CR.13, the reduced models are solved exactly but initial original-space gaps range from 27/640 to 67/100. None is mislabeled as meeting tolerance. One refinement to original singletons closes every gap to zero. This demonstrates both the need to account for aggregation error and the constructive route back to the original problem; it does not claim that one refinement always suffices.

New regression tests include 72 exhaustive original/reduced optimum pairs, 602 fixed-book lifts, 441 cap-dispersion checks, 216 accuracy-binned partitions, and 56 exact refinement comparisons. Genuine clipping at saturation restores every individual cap. The independent checker rejects 18 semantic mutations, including changed representatives, lost branches, altered charges, false gaps, and missing coverage leaves. Prior tests and derivations are preserved without relabeling them as new runs. The study establishes algorithmic and certificate behavior on these declared inputs; it does not assert field calibration or universal runtime superiority.

CR.7 Joint Computational Evidence

The evidence now separates two questions. The preserved distinct-history study tests certified joint design against unaggregated boxes, priced-prefix search, all-book enumeration, and direct mixed-integer envelopes. Its 24 original-space successes at tolerance 0.001, including up to 256 distinct histories, remain in Tables CR.6–CR.7. Those clustered-input results are not a universal runtime claim. The full protocol, four deliberately unresolved coarse stress cases, their refinements, and the earlier R46 failures remain in the computational record and companion tables.

The new paired study isolates harmonic pooling under substantially dispersed costs. It contains 24 cases: twelve primary cases with four, six, or eight histories and twelve cases with 32, 128, or 256 histories. Catalog

sizes are seven and nine, budgets range from one to three, and the four families vary common or dispersed caps, eligibility, saturated or interior promises, positive charges, and a zero-cost history. Within each shared eligibility prefix and cap bin, all curvatures enter one deliberately coarse group; both representatives use identical groups and a minimum-cap guard. Each receives 31 target boxes, four price refinements and reduced tolerance 0.0005. The requested original tolerance is 0.001.

Table CR.12 reports 14 harmonic certificates within tolerance, versus 11 for minimum-curvature aggregation. All 10 unresolved harmonic cases remain in the record; the largest gap is 0.0316. At the twelve primary all-book references, harmonic reduced optima are never larger and are strictly smaller in 5 cases. Nevertheless, a tighter relaxation need not improve every interrupted interval: cost pooling can leave cap dispersion dominant, and independently chosen price bounds can differ. We report both intervals and their valid intersection, not a fictitious pointwise runtime dominance.

All 48 paired upper/lower certificates pass independent semantic checking. The 24 direct tangent/secant MIP solves retain original support probabilities and service decisions; every solver status, time, node count, primal–dual gap, envelope error, and bracket width is reported in Table CR.2. These numerical intervals are not rational proofs. Exact regression checks comprise 32 ordered joint-optimum comparisons, 199 original-policy lifts, 65 dispersion checks, 96 accuracy partitions, 32 refinement comparisons, and 14 rejected semantic mutations. The study tests the new mathematical mechanism rather than replacing unresolved cases with selected successes. No operational population, charge calibration, or deployment is claimed from synthetic inputs.

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CR.8 Complete Paired Outcomes

All cases in the declared protocol are listed. H and M denote harmonic and minimum-curvature aggregation. C and U denote a certified tolerance and an unresolved original-space gap. Times include reduction and lift but exclude independent checking. The objective scale is $r + \Gamma/2$ plus the largest possible selected charge; scaled gaps divide by that stated scale, not by a possibly zero or negative optimum.

Table CR.1: Every original-space interval and execution cost.

Case	k	d	m	H gap	M gap	H sec.	M sec.	H/M	H nodes	Scale
case-00	4	2	1	7.273e-05	0.005223	0.007067	0.01756	C/U	1	3.303
case-01	6	3	3	0.01757	0.01798	0.09332	0.1154	U/U	23	3.511
case-02	8	4	2	0.001003	0.008414	0.008683	0.005686	U/U	1	6.004
case-03	4	4	3	2.196e-05	2.196e-05	0.0151	0.01344	C/C	1	6.044
case-04	6	2	1	0	0.005836	0.006112	0.005672	C/U	1	3.486
case-05	8	3	3	0.02529	0.02522	0.1212	0.093	U/U	31	3.628
case-06	4	4	2	0	0	0.007206	0.005672	C/C	1	6.004
case-07	6	4	3	7.878e-06	7.878e-06	0.0155	0.0134	C/C	1	6.048
case-08	8	2	1	0.0005123	0.009995	0.008584	0.02171	C/U	1	3.593
case-09	4	3	3	0.0008874	0.0008874	0.12	0.1181	C/C	31	3.333
case-10	6	4	2	0.003184	0.009106	0.008	0.005815	U/U	1	6.003
case-11	8	4	3	0.0005434	0.0004811	0.1744	0.1697	C/C	31	6.044
case-12	32	2	1	1.242e-05	0.0001593	0.01876	0.008649	C/C	1	3.884
case-13	32	4	3	0.02647	0.02651	0.2471	0.1906	U/U	31	3.929
case-14	32	4	2	0.002726	0.004005	0.02676	0.009981	U/U	1	6.005
case-15	32	4	3	0.0004471	3.402e-05	0.03163	0.02185	C/C	1	6.05
case-16	128	2	1	1.152e-05	0.0001534	0.08403	0.01338	C/C	1	3.97
case-17	128	4	3	0.02906	0.02903	0.4054	0.1266	U/U	31	4.015
case-18	128	4	2	0.003184	0.004612	0.2144	0.0153	U/U	1	6.005
case-19	128	4	3	4.955e-05	5.621e-05	0.06063	0.02612	C/C	1	6.02
case-20	256	2	1	4.111e-06	0.0001508	0.3307	0.01909	C/C	1	3.985
case-21	256	4	3	0.0316	0.03165	1.095	0.1777	U/U	29	4.025
case-22	256	4	2	0.003552	0.004935	0.7878	0.02204	U/U	1	6.005
case-23	256	4	3	0	7.392e-07	0.141	0.02952	C/C	1	6.025

CR.9 Independent Numerical Mixed-Integer Comparisons

Every primary case has tangent and secant solves of the uneliminated original model. Status 0 means completed; status 1 means a solver limit. Times, nodes and relative primal–dual gaps below are tangent/secant

pairs. The approximation bound is the maximum one-sided quadratic-envelope error. Width is the tangent dual upper bound minus the secant feasible original payoff. These are floating-point diagnostics, never substitutes for the exact semantic certificates.

Table CR.2: All direct MIP comparisons; no timeout or failed solve is omitted.

Case	Status	Seconds	Nodes	MIP gaps	Env. error	Width
case-00	0/0	0.002927/0.00213	1/1	0/0	0.0001154	5.226e-05
case-01	0/0	0.008477/0.007788	1/1	0/0	6.078e-05	0
case-02	0/0	0.02646/0.02367	1/1	0/0	0.00014	6.545e-05
case-03	0/0	0.005721/0.005395	1/1	0/0	0.0001162	0
case-04	0/0	0.002528/0.002488	1/1	0/0	9.626e-05	4.048e-05
case-05	0/0	0.005747/0.005584	1/1	0/0	9.181e-05	0
case-06	0/0	0.008634/0.009637	1/1	0/0	0.0001127	1.759e-05
case-07	0/0	0.02879/0.0256	2/1	0/0	8.268e-05	-3.109e-14
case-08	0/0	0.003193/0.003005	1/1	0/0	0.0001198	9.469e-05
case-09	0/0	0.007799/0.008086	1/1	0/0	7.96e-05	0
case-10	0/0	0.02637/0.02744	1/1	0/0	9.436e-05	1.515e-05
case-11	0/0	0.01151/0.01303	1/1	0/0	9.175e-05	1.11e-16

CR.10 Interpretation and Reproduction

The twelve all-book references independently enumerate menu choices but share the exact fixed-book allocator. The separate MIP retains support indicators, probabilities, pre-draw service and every realized ceiling. The harmonic checker imports neither harmonic reduction nor the search implementation: it rebuilds harmonic means, verifies eligibility and the anchor guard, checks all original lottery constraints, verifies clipping comparison identities and group supporting prices, and delegates the full target-cover and Bellman upper witness to the antecedent independent checker.

The paired study deliberately uses one wide curvature bin within each cap and eligibility group. It is an ablation of the optimistic representative, not an experiment claiming that conservative accuracy bins are always small. The supplied records separately count both accuracy partitions at three tolerances. The final metric pass uses the antecedent node-accounting MIP wrapper; the cases and limits did not change from the initial authoring run. Both local and remote executions must identify their own measured times rather than substitute one environment's times for another's. Full rational policies, raw solver messages, scaled-gap denominators, check times, protocol hashes and generators are retained in the revision directory.

Table CR.3: Theorem-level contribution map. Antecedent statements and proofs remain in the current readers.

Component	Scope	Provenance
Canonical and continuous design	Exact response, institutional endpoints, continuous and Monge results	Antecedent, Theorems 2.1, E.6
Saturated charged path	Globally exact forced targets	R42; Corollary 3.2
Unrestricted price path and prefix search	Polynomial oracle and finite joint search	R43; Theorems B.1–B.2
Conditional compression and full net	Fixed targets; fixed-history additive scheme	R45; Theorems 3.1, A.2
Exact types, two-sided paths and box covers	Exact repeated-type reduction; polynomial box oracle; finite accuracy	R46; Proposition 5.1, Theorems 6.1, 7.1
Guarded response coarsening	Different caps and costs; explicit original-contract loss and lift	R47, Theorem EC.15.1
Variable-history additive scheme	Bounded eligibility and primitive scales; explicit accuracy dependence	R47, Theorem EC.15.3
Certified refinement and checking	Nested valid intervals; original policy, grouping, Bellman and cover checks	R47 refinement, Proposition EC.15.4; extended checker
Harmonic pooling and square-root bins	Dominating upper model; optimal original lift; smaller accuracy-dependent dimension	R48, Theorems EC.16.1–EC.16.4

Table CR.4: Joint design at absolute tolerance 0.001. Counts include every frozen holdout case. Both methods optimize targets and books; neither column is conditional optimization.

Branches	Catalog	Cases	Box completed	Prefix completed	Box seconds	Box max gap
2	9	4	4	4	0.018	0.00057
2	17	4	4	4	0.057	0.00010
4	9	4	4	4	0.028	0.00023
4	17	4	4	4	0.093	0.00094
8	9	4	3	4	0.106	0.01248
8	17	4	2	4	4.103	0.00673
128	33	4	3	4	0.398	0.00229
256	33	2	2	2	0.306	0.00078
16	33	2	2	2	0.850	0.00075

Seconds are medians for box optimization, excluding independent verification. The price-prefix baseline uses the inherited R43 completion bound, not merely an unpriced catalog relaxation. The 128- and 256-branch rows use exactly repeated types; the 128-branch row contains both four- and eight-type cases. Node and time limits, and all individual outcomes, are reported in the companion and machine-readable records.

Table CR.5: Completed and interrupted target grids versus adaptive joint design on identical finite catalogs. All have $N = 5$ and $m = 2$. A dash means that the grid does not certify the requested accuracy.

Branches	Tolerance	Grid profiles	Grid seconds	Grid certified	Box nodes	Box gap
2	0.100	7	0.005	Yes	1	0.00000
2	0.010	54	0.033	Yes	1	0.00000
2	0.001	527	0.311	Yes	1	0.00000
3	0.100	1000	0.826	–	1	0.00003
3	0.010	1000	0.904	–	1	0.00003
3	0.001	1000	0.908	–	1	0.00003
4	0.100	1000	1.006	–	1	0.01354
4	0.010	1000	1.063	–	3	0.00903
4	0.001	1000	1.086	–	1001	0.00128
5	0.100	1000	1.192	–	1	0.00001
5	0.010	1000	1.163	–	1	0.00001
5	0.001	1000	1.172	–	1	0.00001

The grid has a declared cap of 1,000 evaluated profiles; these are profile-limited comparisons, not claims of time-matched speedup. Adaptive runs use at most 1,001 nodes and an eight-second soft limit. Exact values, objective scales, losses, and runtimes for both algorithms appear in the archived records.

Table CR.6: Distinct-history joint design at original-space tolerance 0.001. Every declared case is included. Histories in these rows have distinct cap–cost pairs.

Histories	Catalog	Cases	Coarsened completed	Raw boxes completed	Priced prefix completed	Groups
4	9	4	4	4	4	4
8	9	4	4	2	4	5
12	9	4	4	4	4	5
32	17	4	4	3	4	5
128	17	4	4	3	3	5
256	17	4	4	3	2	5

The twelve primary cases use four, eight, or twelve histories, nine catalog levels, and two or three commands. The twelve scaling cases use seventeen catalog levels and three commands. The reduced solve uses tolerance 0.0005; completion above refers only to the checked original interval. Both box methods use at most 101 nodes and a four-second soft limit checked after full nodes. The priced-prefix comparator has the same requested original tolerance, node limit, and soft limit. All raw intervals and all lifted intervals pass their independent checkers.

Table CR.7: Optimization time and original-space gaps on all distinct-history cases. Seconds are local medians; maximum gaps are exact certificate widths rounded for display.

Histories	Coarsened seconds	Raw box seconds	Prefix seconds	Coarsened max gap	Raw box max gap	Prefix max gap
4	0.027	0.023	0.037	0.00034	0.00034	0.00081
8	0.246	0.684	0.069	0.00052	0.00212	0.00092
12	0.010	0.019	0.017	0.00010	0.00000	0.00000
32	0.618	0.372	0.315	0.00078	0.00261	0.00070
128	0.049	1.308	1.123	0.00013	0.00965	0.00965
256	0.095	3.981	3.503	0.00006	0.00173	0.00173

Timing excludes independent checking; coarsened timing includes reduction and original-space lifting. It is not uniformly faster: per-case outcomes, actual times above the soft limit, node counts, objective scales, and normalized gaps are retained in `results/joint.json`. The current code uses exact rational arithmetic. The data are a declared four-cluster perturbation family, not arbitrary-population or field evidence.

Table CR.8: Original-budget comparison across six seeds

Charges	Cases	Core exact	Safeguarded exact in sample	Core wins	Greedy wins	Max core loss
None	18	18	18	2	0	0.000000
Nonuniform	18	17	18	8	1	0.028056

Exact means equality with the exhaustive joint catalog reference. Core and greedy wins compare their same-budget values; ties are omitted. The safeguarded policy keeps the better feasible candidate. The charged exception remains in the data. These counts are observations, not global guarantees for a finite target portfolio.

Table CR.9: Completed same-budget target-net comparisons

Branches	Tolerance	Cases	Profiles	Maximum actual loss
2	1/4	4	28	0.000000
2	1/8	4	50	0.000000
2	1/16	4	96	0.000000
3	1/2	4	47	0.049805
3	1/4	4	164	0.034645

All completed nets meet the requested additive tolerance with the original two-symbol budget and nonuniform charges. Counts aggregate four seeds. An additional interrupted run withholds its global accuracy guarantee.

Table CR.10: Conditional frontier scaling (targets fixed) with heterogeneous realization ceilings

Seed	Branches	Catalog	Budget	Charges	Selected	Seconds	Gap
104729	32	17	4	No	4	0.036	0.019173
104729	32	17	4	Yes	4	0.036	0.037009
104729	128	33	8	No	8	0.486	0.004410
104729	128	33	8	Yes	5	0.484	0.041534
104729	256	65	8	No	8	3.257	0.004923
104729	256	65	8	Yes	8	3.278	0.025671
205019	32	17	4	No	4	0.038	0.010627
205019	32	17	4	Yes	3	0.038	0.040935
205019	128	33	8	No	8	0.474	0.003661
205019	128	33	8	Yes	5	0.478	0.042000
205019	256	65	8	No	8	3.303	0.004099
205019	256	65	8	Yes	7	3.316	0.023976

Seconds are medians of three repeated executions on the same input. Gap is the valid same-budget Jensen upper bound minus the ideal-target conditional policy value, not an observed error to an unknown global optimum. Full timing arrays, stored-entry bit lengths and cumulative process resident-memory records are archived.

Table CR.11: Adaptive candidates at an unchanged installed-symbol budget

Case	Candidate pool	Candidates	Symbols	Exact-reference loss
interior-ideal	uniform	9	2	0.044030
interior-ideal	uniform	65	2	0.002991
interior-ideal	oracle pool	4	2	0.000000
off-cap-top	uniform	9	2	0.000000
off-cap-top	Adaptive	5	2	0.000781
off-cap-top	Adaptive	6	2	0.000000
Sat. 104729	uniform	9	3	0.006507
Sat. 104729	Adaptive	9	3	0.000000
Sat. 205019	uniform	9	3	0.001824
Sat. 205019	Adaptive	9	3	0.000000

Every row has an exact continuous reference in the stated Jensen-attaining or saturated expected-participation regime. The interior oracle pool explicitly inserts ideal targets and is a sanity check. Candidate-pool size is not installed memory. Adaptive midpoint selection is a heuristic; catalog optimization at the saturated target is exact. The complete 22-row path is in the companion and JSON.

Table CR.12: Harmonic versus minimum-curvature aggregation at identical partitions and search limits. Entries report original-space certificates at tolerance 0.001, not reduced-solver completion. Every family contains six instances, including 32, 128, and 256 histories.

Family	Certified / 6		Maximum gap		Median seconds	
	Harmonic	Minimum	Harmonic	Minimum	Harmonic	Minimum
Common caps	6	3	0.0005123	0.009995	0.01367	0.01547
Dispersed caps	1	1	0.0316	0.03165	0.1842	0.1223
Eligibility / high promise	1	1	0.003552	0.009106	0.01772	0.007898
Zero cost / charged	6	6	0.0005434	0.0004811	0.04613	0.02399

Both methods receive 31 target boxes and four price refinements. Larger exact rational denominators and optimal original-group recovery can make harmonic pooling slower. A tighter reduced optimum is not a theorem about interrupted-search runtime or every returned interval.

Table CR.13: Coarse-relaxation stress cases. The root obligation equals total original capacity. Each initial two-group solve is exact for its reduced model but does not certify the original tolerance.

Case	Commands	Final curvature	Original optimum	Initial lower	Initial gap	Refined gap
0	2	2.5	0.86781	0.86781	0.04219	0.00000
1	2	6.0	0.86562	0.86562	0.04437	0.00000
2	1	2.5	-0.45200	-0.45200	0.11000	0.00000
3	1	6.0	-1.06200	-1.06200	0.67000	0.00000

Caps are $(1/5, 2/5, 4/5)$, weights $(1/5, 3/10, 1/2)$, and ceilings are all one. The first two curvatures are one and two; the third is shown. The catalog is $\{0, 1/4, 1/2, 3/4, 1\}$. All charges are zero except case 3, which charges $1/20$ per installed level. The initial groups are $\{1\}$ and $\{2, 3\}$, with the smallest-cap guard intact. Splitting the latter group gives three singleton histories and zero global gap. Every stage has a checked original-space policy and upper certificate.

Table CR.14: Independent original-model mixed-integer comparisons on every primary case. Times and node counts are tangent/secant pairs. All widths are numerical, not rational certificates.

Case	Seconds	Nodes	Max solver gap	Envelope error	Signed width	Status pair
00	0.02/0.01	1/1	0.0e+00	1.0e-04	1.9e-05	0/0
01	0.28/0.23	1/1	0.0e+00	7.7e-05	-6.7e-16	0/0
02	0.02/0.02	1/1	0.0e+00	7.1e-05	1.7e-05	0/0
03	0.02/0.05	1/1	0.0e+00	7.4e-05	1.8e-15	0/0
04	0.02/0.02	1/1	0.0e+00	7.2e-05	2.7e-05	0/0
05	0.07/0.07	1/1	0.0e+00	7.9e-05	-2.2e-16	0/0
06	0.01/0.03	1/1	0.0e+00	7.3e-05	4.3e-06	0/0
07	0.26/0.14	1/1	0.0e+00	7.7e-05	2.2e-16	0/0
08	0.02/0.02	1/1	5.6e-08	5.6e-05	1.3e-05	0/0
09	0.04/0.03	1/1	0.0e+00	8.5e-05	0.0e+00	0/0
10	0.01/0.01	1/1	0.0e+00	8.0e-05	1.1e-16	0/0
11	0.18/0.22	1/1	0.0e+00	6.7e-05	0.0e+00	0/0

Each envelope has 64 segments and a separate two-second solver limit. Status 0 means the numerical solver reported optimal termination; nonzero statuses would remain in the table. All 24 solves report status 0 here. Signed negative widths at floating roundoff scale are deliberately retained. The full formulation, global numerical bounds, primal values, residuals, probabilities, and individual runtimes are in `results/mip.json`. Exhaustive book references agree with these numerical brackets up to the declared 10^{-6} comparison allowance, which is not an exact upper-bound guarantee.

Table CR.15: Individual joint certificates. The first column is the reproducible instance identifier.

Instance	Types	Nodes	Seconds	Lower	Upper	Gap/scale
h2-9-0-0	2	1	0.020	0.79453	0.79511	0.00010
h2-9-0-1	2	1	0.014	0.57578	0.57578	0.00000
h2-9-1-0	2	1	0.018	0.82008	0.82053	0.00007
h2-9-1-1	2	1	0.018	0.83589	0.83595	0.00001
h2-17-0-0	2	1	0.078	0.66865	0.66875	0.00002
h2-17-0-1	2	1	0.042	0.65984	0.65984	0.00000
h2-17-1-0	2	1	0.052	0.71419	0.71429	0.00002
h2-17-1-1	2	1	0.063	1.11961	1.11961	0.00000
h4-9-0-0	4	1	0.030	0.81641	0.81663	0.00004
h4-9-0-1	4	1	0.025	0.64713	0.64713	0.00000
h4-9-1-0	4	1	0.027	0.50484	0.50484	0.00000
h4-9-1-1	4	1	0.030	0.42283	0.42283	0.00000
h4-17-0-0	4	1	0.077	0.49392	0.49412	0.00003
h4-17-0-1	4	1	0.094	0.58137	0.58219	0.00014
h4-17-1-0	4	99	1.943	0.56328	0.56422	0.00015
h4-17-1-1	4	1	0.092	0.65512	0.65535	0.00004
h8-9-0-0	8	1	0.046	0.60338	0.60339	0.00000
h8-9-0-1	8	7	0.160	0.81156	0.81170	0.00002
h8-9-1-0	7	301	4.437	0.64127	0.65375	0.00203
h8-9-1-1	8	1	0.051	0.66771	0.66771	0.00000
h8-17-0-0	8	169	8.047	0.65177	0.65394	0.00036
h8-17-0-1	8	1	0.156	0.62026	0.62116	0.00015
h8-17-1-0	8	201	8.052	0.43832	0.44505	0.00109
h8-17-1-1	8	1	0.158	0.88753	0.88770	0.00003
s4-33-32-0	4	1	0.276	0.57796	0.57874	0.00013
s4-33-32-1	4	1	0.331	0.67633	0.67635	0.00000
s4-33-64-0	4	1	0.274	0.57796	0.57874	0.00013
s4-33-64-1	4	1	0.337	0.67633	0.67635	0.00000
s8-33-16-0	8	1	0.466	0.63893	0.63894	0.00000
s8-33-16-1	8	43	8.306	0.76754	0.76983	0.00037
s16-33-1-0	16	1	0.867	0.54159	0.54233	0.00012
s16-33-1-1	16	1	0.833	0.74255	0.74315	0.00010

Table CR.16: Independent numerical MIP comparisons for all 24 primary holdout cases. Times and node counts are tangent/secant pairs. Widths are numerical, not rational certificates.

Instance	Seconds	Nodes	Max MIP gap	Envelope	Width	Prefix seconds
h2-9-0-0	0.03/0.02	1/1	0.0e+00	8.4e-05	0.0e+00	0.015
h2-9-0-1	0.01/0.01	1/1	0.0e+00	4.2e-05	0.0e+00	0.027
h2-9-1-0	0.01/0.01	1/1	0.0e+00	8.6e-05	0.0e+00	0.018
h2-9-1-1	0.01/0.00	1/1	0.0e+00	3.1e-05	0.0e+00	0.022
h2-17-0-0	0.02/0.02	1/1	0.0e+00	5.6e-05	0.0e+00	0.042
h2-17-0-1	0.02/0.02	1/1	0.0e+00	2.8e-05	0.0e+00	0.043
h2-17-1-0	0.01/0.00	1/1	0.0e+00	3.3e-05	1.1e-16	0.040
h2-17-1-1	0.01/0.00	1/1	0.0e+00	1.7e-04	2.2e-16	0.056
h4-9-0-0	0.01/0.02	1/1	0.0e+00	4.7e-05	0.0e+00	0.024
h4-9-0-1	0.01/0.01	1/1	0.0e+00	2.1e-05	0.0e+00	0.026
h4-9-1-0	0.01/0.00	1/1	2.2e-16	1.5e-05	1.1e-16	0.022
h4-9-1-1	0.01/0.01	1/1	0.0e+00	1.2e-05	4.8e-06	0.027
h4-17-0-0	0.03/0.02	1/1	0.0e+00	1.7e-05	1.7e-16	0.065
h4-17-0-1	0.03/0.02	1/1	0.0e+00	2.6e-05	1.1e-16	0.120
h4-17-1-0	0.03/0.03	1/1	0.0e+00	3.4e-05	0.0e+00	0.087
h4-17-1-1	0.01/0.01	1/1	0.0e+00	1.8e-05	0.0e+00	0.125
h8-9-0-0	0.02/0.01	1/1	0.0e+00	6.4e-05	8.5e-05	0.041
h8-9-0-1	0.05/0.05	1/1	0.0e+00	3.5e-05	5.3e-15	0.075
h8-9-1-0	0.04/0.04	1/1	0.0e+00	6.3e-05	0.0e+00	0.063
h8-9-1-1	0.01/0.01	1/1	0.0e+00	3.4e-05	4.1e-05	0.045
h8-17-0-0	0.12/0.11	1/1	0.0e+00	5.8e-05	0.0e+00	0.179
h8-17-0-1	0.03/0.03	1/1	0.0e+00	3.6e-05	0.0e+00	0.123
h8-17-1-0	0.01/0.01	1/1	1.3e-16	1.5e-05	5.6e-17	0.170
h8-17-1-1	0.02/0.02	1/1	0.0e+00	4.9e-05	2.2e-06	0.128

Each envelope solve has a four-second limit and 64 shortfall segments. All 48 solves returned optimal status in this execution; hence no timeout is hidden. Maximum mixed-integer primal–dual gaps and the per-case shortfall-envelope errors are reported above. Signed raw widths, including negative values at floating roundoff scale, solver messages, residuals, solutions, and separate node counts are retained in `comparators.json`. Display widths truncate only negative floating residuals to zero; this is not a rational exactness assertion. The companion also retains the separate eight-case initial MIP execution, which is not pooled into the 24-case comparison.

Table CR.17: Accounting for Theorem E.6; ordered rational input, $1 \leq m \leq k$.

Cost component	Included work or storage
Input validation and six prefix moments	$O(k)$ rational operations; $O(k)$ stored values.
Sorting	Input interface requires ordered distinct caps; otherwise add $O(k \log k)$ exact comparisons and reorder associated primitives.
Continuous terminal minimization	Included in $O(k)$ oracle requests; each interval minimizes its quadratic exactly, including singleton and endpoint cases.
Key comparisons and repeated requests	Included in $O(mk)$ work; a repeated finite request is charged again. No quadratic entry cache is assumed.
Padding comparisons	Included index comparisons; undefined economic costs are never evaluated.
Previous-layer values and backpointers	$O(mk)$ stored values and indices, plus $O(k)$ working search space.
Reconstruction and output	One winning codebook per budget; $O(m^2) \leq O(mk)$ symbols written. No reconstruction at every candidate endpoint.
Rational heights and integer operations	$O(S)$ bits per evaluated rational, $S = O(k[L + \log(k+2)])$; conservative $O(mkS^3)$ bit operations, including reduction and comparison.
Physical bit storage	$O((mk + k)[S + \log(k+2)])$ bits for the stated arrays and indices; not a claim of $O(mk)$ bits.
Inherited public graph and response compiler	Not invoked by the renewal algorithm; their read-only data, compilation, and query costs enter the separate implementation menu when used.
External solver preprocessing	Not part of the frontier theorem. Finite-network construction, floating solver work, and exact certification are measured separately.

CR.11 Complete Exact-Pooling Records

P00–P23 identify the unchanged paired cases. $Ck/N/s$ identifies history count, catalog size, and seed offset from 49400. $Ss/a/n$ identifies seed offset from 49500, inverse tolerance, and node allowance. These are labels for the full machine-readable identifiers, not additional cases. Times below are measured in the execution environment recorded in `results/study.json`. Optimization and independent checking are timed separately.

The scale is $A = r + \max_j \gamma_j/2 + \sum_{i=1}^m \rho(i)$, where the charges are sorted from largest to smallest. The normalized interval width is gap divided by A . This is a scale diagnostic, not a relative-error theorem. An exact-reference loss is reported only when a reference was specified; a dash means that no exhaustive reference was requested. All common-eligibility rows instead have exact zero-width certificates.

Table CR.18: All 64 exact-pooling runs, with interruptions retained.

Identifier	Classes	Nodes	Gap	Seconds	Check seconds	Complete
P00	1	1	0	0.0031	0.0018	Yes
P01	1	1	0	0.0041	0.0028	Yes
P02	3	1	0	0.0247	0.0046	Yes
P03	3	1	0.00002196	0.024	0.003	Yes
P04	1	1	0	0.0039	0.0024	Yes
P05	1	1	0	0.0052	0.0035	Yes
P06	3	1	0	0.017	0.0027	Yes
P07	3	1	0.00000788	0.0346	0.0041	Yes
P08	1	1	0	0.0051	0.0032	Yes
P09	1	1	0	0.0031	0.0022	Yes
P10	3	1	0	0.0201	0.0038	Yes
P11	3	1	0.00001723	0.041	0.0052	Yes
P12	1	1	0	0.0253	0.0177	Yes
P13	1	1	0	0.0239	0.0188	Yes
P14	3	1	0	0.1349	0.0219	Yes
P15	3	5	0.00013796	0.509	0.1241	Yes
P16	1	1	0	0.1598	0.1225	Yes
P17	1	1	0	0.1113	0.0919	Yes
P18	3	1	0	0.9237	0.0841	Yes
P19	3	1	0.00008945	0.743	0.1133	Yes
P20	1	1	0	0.634	0.5141	Yes
P21	1	1	0	0.2691	0.2441	Yes
P22	3	1	0	2.9379	0.1681	Yes
P23	3	1	0	1.5381	0.2627	Yes

Identifier	Classes	Nodes	Gap	Seconds	Check seconds	Complete
C16/9/0	1	1	0	0.0123	0.0088	Yes
C16/9/1	1	1	0	0.0123	0.009	Yes
C16/17/0	1	1	0	0.0278	0.0202	Yes
C16/17/1	1	1	0	0.0279	0.0202	Yes
C64/9/0	1	1	0	0.0431	0.0305	Yes
C64/9/1	1	1	0	0.0435	0.0307	Yes
C64/17/0	1	1	0	0.0983	0.0724	Yes
C64/17/1	1	1	0	0.0985	0.072	Yes
C256/9/0	1	1	0	0.1684	0.121	Yes
C256/9/1	1	1	0	0.171	0.1233	Yes
C256/17/0	1	1	0	0.3831	0.2866	Yes
C256/17/1	1	1	0	0.3838	0.2921	Yes
C1024/9/0	1	1	0	0.6825	0.4881	Yes
C1024/9/1	1	1	0	0.6852	0.4852	Yes
C1024/17/0	1	1	0	1.5299	1.1907	Yes
C1024/17/1	1	1	0	1.5764	1.166	Yes
S0/100/31	6	1	0.00333333	0.0619	0.0064	Yes
S0/100/127	6	1	0.00333333	0.0612	0.0064	Yes
S0/1000/31	6	31	0.00291667	0.7284	0.1873	No
S0/1000/127	6	127	0.00177083	2.8245	0.7487	No
S1/100/31	7	3	0.00442073	0.1334	0.0207	Yes
S1/100/127	7	3	0.00442073	0.129	0.021	Yes
S1/1000/31	7	31	0.00196265	0.8466	0.2089	No
S1/1000/127	7	127	0.0012862	3.1916	0.8452	No
S2/100/31	6	1	0.00299609	0.0634	0.0089	Yes
S2/100/127	6	1	0.00299609	0.0898	0.0095	Yes
S2/1000/31	6	31	0.00223437	0.8271	0.208	No
S2/1000/127	6	127	0.00099658	3.3428	0.8682	Yes
S3/100/31	8	1	0.00201231	0.0501	0.0053	Yes
S3/100/127	8	1	0.00201231	0.0505	0.0052	Yes
S3/1000/31	8	31	0.00176077	0.5222	0.1235	No
S3/1000/127	8	127	0.00150923	1.9785	0.4868	No
S4/100/31	5	1	0.00285156	0.0619	0.007	Yes
S4/100/127	5	1	0.00285156	0.0627	0.007	Yes
S4/1000/31	5	31	0.00122446	0.8193	0.2089	No
S4/1000/127	5	55	0.00091647	1.4309	0.4059	Yes
S5/100/31	6	13	0.00977539	0.3649	0.0834	Yes
S5/100/127	6	13	0.00977539	0.3619	0.0837	Yes

Identifier	Classes	Nodes	Gap	Seconds	Check seconds	Complete
S5/1000/31	6	31	0.00904297	0.7993	0.2028	No
S5/1000/127	6	127	0.00590576	3.1442	0.8246	No

Table CR.19: Scale diagnostics and exact-reference incumbent losses; no missing reference is imputed.

Identifier	Objective scale	Gap / scale	Reference loss	Relative loss
P00	3.3025	0	0	0
P01	3.510969	0	0	0
P02	6.004444	0	0	0
P03	6.044	0.00000363	0	0
P04	3.485969	0	0	0
P05	3.628364	0	0	0
P06	6.004	0	0	0
P07	6.047857	0.0000013	0	0
P08	3.593364	0	0	0
P09	3.3325	0	0	0
P10	6.002857	0	0	0
P11	6.044444	0.00000285	0	0
P12	3.884355	0	—	—
P13	3.929355	0	—	—
P14	6.004848	0	—	—
P15	6.049848	0.0000228	—	—
P16	3.970078	0	—	—
P17	4.015078	0	—	—
P18	6.004961	0	—	—
P19	6.019961	0.00001486	—	—
P20	3.984952	0	—	—
P21	4.024952	0	—	—
P22	6.004981	0	—	—
P23	6.024981	0	—	—
C16/9/0	8.12	0	—	—
C16/9/1	8.16	0	—	—
C16/17/0	8.16	0	—	—
C16/17/1	8.18	0	—	—
C64/9/0	8.18	0	—	—
C64/9/1	8.16	0	—	—

Identifier	Objective scale	Gap / scale	Reference loss	Relative loss
C64/17/0	8.18	0	–	–
C64/17/1	8.18	0	–	–
C256/9/0	8.15	0	–	–
C256/9/1	8.16	0	–	–
C256/17/0	8.18	0	–	–
C256/17/1	8.17	0	–	–
C1024/9/0	8.16	0	–	–
C1024/9/1	8.17	0	–	–
C1024/17/0	8.16	0	–	–
C1024/17/1	8.18	0	–	–
S0/100/31	4.22	0.00078989	0	0
S0/100/127	4.22	0.00078989	0	0
S0/1000/31	4.22	0.00069115	0	0
S0/1000/127	4.22	0.00041963	0	0
S1/100/31	3.95	0.00111917	0	0
S1/100/127	3.95	0.00111917	0	0
S1/1000/31	3.95	0.00049687	0	0
S1/1000/127	3.95	0.00032562	0	0
S2/100/31	4.115	0.00072809	0	0
S2/100/127	4.115	0.00072809	0	0
S2/1000/31	4.115	0.00054298	0	0
S2/1000/127	4.115	0.00024218	0	0
S3/100/31	4.115	0.00048902	0	0
S3/100/127	4.115	0.00048902	0	0
S3/1000/31	4.115	0.00042789	0	0
S3/1000/127	4.115	0.00036676	0	0
S4/100/31	4.035	0.00070671	0	0
S4/100/127	4.035	0.00070671	0	0
S4/1000/31	4.035	0.00030346	0	0
S4/1000/127	4.035	0.00022713	0	0
S5/100/31	3.99	0.00244997	0	0
S5/100/127	3.99	0.00244997	0	0
S5/1000/31	3.99	0.00226641	0	0
S5/1000/127	3.99	0.00148014	0	0

Table CR.20: Every direct mixed-integer solve. Bounds and solver gaps are numerical, not exact certificates.

Case	Envelope	Status	Seconds	Nodes	Solver gap	Error	Bracket
P00	Tangent	0	0.0028	1	0	0.00011545	0.00005226
P00	Secant	0	0.0021	1	0	0.00011545	0.00005226
P01	Tangent	0	0.0083	1	0	0.00006078	0
P01	Secant	0	0.0078	1	0	0.00006078	0
P02	Tangent	0	0.0258	1	0	0.00013997	0.00006545
P02	Secant	0	0.0229	1	0	0.00013997	0.00006545
P03	Tangent	0	0.0055	1	0	0.00011618	0
P03	Secant	0	0.0053	1	0	0.00011618	0
P04	Tangent	0	0.0025	1	0	0.00009626	0.00004048
P04	Secant	0	0.0025	1	0	0.00009626	0.00004048
P05	Tangent	0	0.0057	1	0	0.00009181	0
P05	Secant	0	0.0055	1	0	0.00009181	0
P06	Tangent	0	0.0085	1	0	0.0001127	0.00001759
P06	Secant	0	0.0093	1	0	0.0001127	0.00001759
P07	Tangent	0	0.0281	2	0	0.00008268	-3.1e-14
P07	Secant	0	0.0248	1	0	0.00008268	-3.1e-14
P08	Tangent	0	0.0031	1	0	0.00011981	0.00009469
P08	Secant	0	0.0029	1	0	0.00011981	0.00009469
P09	Tangent	0	0.0077	1	0	0.0000796	0
P09	Secant	0	0.0082	1	0	0.0000796	0
P10	Tangent	0	0.0259	1	0	0.00009436	0.00001515
P10	Secant	0	0.0268	1	0	0.00009436	0.00001515
P11	Tangent	0	0.0111	1	0	0.00009175	1.1e-16
P11	Secant	0	0.0129	1	0	0.00009175	1.1e-16
S0/100/31	Tangent	0	0.0171	1	1.8e-16	0.00011199	0
S0/100/31	Secant	0	0.0147	1	0	0.00011199	0
S1/100/31	Tangent	0	0.0131	1	0	0.00013239	0
S1/100/31	Secant	0	0.0129	1	0	0.00013239	0
S2/100/31	Tangent	0	0.005	1	0	0.0000432	5.6e-17
S2/100/31	Secant	0	0.0067	1	0	0.0000432	5.6e-17
S3/100/31	Tangent	0	0.0173	1	0	0.00005913	-1.4e-17
S3/100/31	Secant	0	0.011	1	0	0.00005913	-1.4e-17
S4/100/31	Tangent	0	0.006	1	1.2e-16	0.00002769	0
S4/100/31	Secant	0	0.006	1	1.2e-16	0.00002769	0
S5/100/31	Tangent	0	0.0079	1	0	0.0000883	0

Case	Envelope	Status	Seconds	Nodes	Solver gap	Error	Bracket
S5/100/31	Secant	0	0.0079	1	0	0.0000883	0

Status zero denotes successful numerical termination. A dash denotes a field the solver did not return, never an inferred zero. Error is the uniform quadratic-envelope allowance; bracket is the tangent upper bound minus the original-payoff value of the secant feasible policy. Small signed roundoff widths, when present, are retained in the raw records. Both envelopes have 32 segments and a three-second limit. The uneliminated probabilities, service tiers, selected levels, model residuals, primal objective and numerical dual bound are retained for every solve.

Historical distinctions. The archived charged portfolio example has exact net value 1373/3600, portfolio value 53/150, and loss 101/3600, approximately 7.36 percent of the positive exact net value. Its safeguarded exactness is a sample observation. The inserted-ideal-target continuous example is an oracle-candidate sanity check. The historical large conditional table explicitly keeps its targets-fixed label. None is reclassified as an exact-pooling result.